New Zealand Felt RAPID Intensity Attenuation Relation Model Data Collection Pipeline

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Method Outline

An earthquake-cell dataset was constructed from the GeoNet earthquake catalogue, GeoNet moment-tensor, GeoNet Felt RAPID, and New Zealand Vs30 data for the period from 1st September 2016 until the 17th October 2026. Catalogue events were acquired in monthly requests for the defined New Zealand region bounding box, with a minimum catalogue magnitude value of 3.0. Catalogue event identifiers, origin times, coordinates, and depths were standardised and combined, with one catalogue row per *publicid* retained.

For each catalogue identifier, Felt RAPID intensity data was requested using median aggregation. The cell identifier, coordinates, median reported intensity, and number of contributing reports were retained. The observations represent reporting cells, not individual responses.

Moment-magnitude (Mw) data was obtained from GeoNet's moment-tensor solutions and matched by *publicid*. Where multiple valid solutions were available, a selection rule prioritising variance reduction, followed by station count, was used. Moment-magnitude was the only magnitude used, catalogue magnitude was not substituted where moment-magnitude was missing. All other fields remained those from the earthquake catalogue. Only observations with Mw greater than or equal to 3.5 were retained. Approximate epicentral and hypocentral distances were calculated.

Vs30 site condition values were extracted from the 2019 Foster et al. Vs30 raster at each unique reporting coordinate. The raster was interpreted in NZTM2000, and reporting points were transformed into it's coordinate system. Extracted values outside a defined range were labelled missing, while observations lacking a Vs30 value were retained.

Pipeline Flow

Script	Function
00_get_geonet_catalogue.R	Retrieve the monthly earthquake catalogue data from GeoNet[1], save as CSV files.
01_combine_geonet_catalgoue.R	Standardise and combine the monthly CSV files into a single catalogue.
02_get_geonet_felt_rapid.R	Retrieve the median reported intensity (Felt RAPID) for each event from GeoNet[2].
03_get_geonet_moment_tensor.R	Match moment-tensor Mw from GeoNet GitHub[3] to events.
04_build_nz_fr_iar_dataset.R	Join events to cells and calculate distances, output dataset.
05_add_vs30_site_conditions.R	Extract Vs30[4] values and join to dataset.

Data is saved in the folder *data_raw,* while outputs for use in analysis are saved in the folder *data_processed*.

1 https://quakesearch.geonet.org.nz/csv?
2 https://api.geonet.org.nz/intensity
3 https://raw.githubusercontent.com/GeoNet/data/main/moment-tensor/GeoNet_CMT_solutions.csv
4 esp4bf02915-sup-0003.tif from https://onlinelibrary.wiley.com/doi/10.1193/121118EQS281M

Primary Units

Unit	Description
Report	One response, which contributes to a GeoNet aggregated intensity.
Cell	The geographic group represented by one returned API feature.
Earthquake-Cell Row	The connection between one earthquake event and one reporting cell.

An earthquake can contribute many rows, and a reporting coordinate can recur for many earthquakes. If an earthquake has reports from 2 cells, one with 2 reports and the other with 20, there are 2 rows and 22 reports for that earthquake. These should NOT be treated as 22 independently located attenuation observations.

GeoNet aggregates Felt RAPID report locations to ensure the privacy of respondents. However, this also creates a limitation as the cell coordinates are representative only.

Identifiers

Identifier	What it Identifies
publicid	Event
publicid + cell_id	Observation
report_longitude + report_latitude	Report Location

Pipeline Development Limitations

A limitation during the development of the data collection pipeline was that all later stages of the project (exploratory data analysis, modelling, and report creation) were dependent on the pipeline. This meant that any changes made to the pipeline would require team members to create downstream changes in their EDA and modelling scripts, as well as to the report. This created a tradeoff which had to be considered, between experimenting with alternatives and updating the pipeline, and ensuring continuity and ease of use for other team members.

As the pipeline always updates the catalogue to the current date (with the default settings), figures generated and recorded at one point in time are likely to have shifted at a later date as more data has become available. To address this, a ‘final’ dataset was created with a set end-date of 17/10/2026.

Script_00 – Earthquake Catalogue

Output

One catalogue CSV for each month within the specified timeframe, saved with the format YYYY_MM.csv to the folder *data_raw/geonet_catalogue*.

Settings/Filters

Setting/Filter	Purpose	Value Used
Minimum Catalogue Magnitude	To exclude earthquakes for which there will almost certainly not be valid Felt RAPID reports or moment-magnitude Mw data.	3.0
Bounding Box	Only download earthquakes within the NZ region.	163.5205,-49.1817,-176.9238,-32.2871[5]
Start Date	The start date from which monthly downloads should begin.	2016-09-01[6]
Overwrite Existing	Avoids re-downloading previously obtained months.	TRUE (for final dataset creation, FALSE during general use)

Main Script Functions

- The beginning of the current month is identified and a sequence of month start dates is generated. The loop uses the next month start date as the end of each completed month's request. For the current month, it uses the date one day after the current date as the request end, or alternatively a specific end date may be set, as was done for this project for data consistency. Midnight (T00:00:00) is appended to the dates when the query is made.
- The query url contains the bounding box, minimum catalogue magnitude, start date and end date. GeoNet QuakeSearch dates are in UTC[7].
- Downloaded files are read and saved as CSV files.

Design Choices & Limitations

- The decision to download earthquake catalogue data for one month at a time was made to avoid exceeding GeoNet's request limit of 20,000 events, confirmed that no month met this limit, so no data was inadvertently lost.
- The decision to use a minimum catalogue magnitude of 3 was made to retain earthquakes which may still be later matched to a moment-magnitude Mw of 3.5 or higher. While ideally all earthquake events of any magnitude would be downloaded, this was impractical in terms of time use and earthquakes below the minimum threshold almost certainly lack moment-magnitude and Felt RAPID report data.
- The start date is the date from which Felt RAPID reports first became available.
- The bounding box used is the default used by GeoNet QuakeSearch to cover the whole of New Zealand.
- A progress message function provides updates on which month is currently being downloaded, and the number of earthquakes downloaded for each month, primarily so the user can see how far along the process is.

5 Taken from https://quakesearch.geonet.org.nz/. The default bounding box coordinates. West, South, East, North.
6 Date from which Felt RAPID reports first became available.
7 https://www.geonet.org.nz/data/faqs

- A pause of 1 second is used between each request as good practice and to avoid being banned/throttled from the API.
- Any change to the minimum catalogue magnitude or bounding box requires a rerun of the script with *overwrite_existing* set to TRUE, in order to update pre-existing monthly CSVs.

Script_01 – Create Combined Catalogue

Output

A combined earthquake catalogue, *geonet_catalogue.parquet* is saved in the *data_raw* folder.

Main Script Functions

- Locates every monthly CSV in data_raw/geonet_catalogue, sorts them by name (which also sorts by date due to the YYYY_MM naming used), reads them, and then writes them to the combined catalogue parquet.
- Recognise missing values, trim whitespace, and standardise column names to lowercase.
- Ensure required columns are present.
- Trim event identifiers, change empty identifier to missing, set UTC explicitly for times, and parse latitude, longitude, and depth as doubles.
- Keep only the first row for each event identifier to avoid an event being joined to each reporting cell multiple times.
- Write output.

Design Choices & Limitations

- Apache Parquet was used via the *arrow* R library as parquet is read faster within R, and also because it avoids R from guessing data types like with CSVs[8].
- Every column is first read as text to avoid issues caused by automatic type inference, where the same column in different files is classified as a different type. Column types are then set explicitely.
- When keeping only the first row for each event identifier, this is a deterministic rule for simplicity sake. The selected occurrence is not necessarily the best catalogue solution. This is a limitation that would require extra research and development to improve.

Script_02 – Felt RAPID

Output

Felt Rapid report data is saved in *data_raw/felt_rapid_observations.parquet*.

Main Script Functions

- Read the combined earthquake catalogue.
- Create an empty tibble containing the required columns, so that queries with no reports still create a consistently formatted output and avoid row-binding errors.
- Request the Felt RAPID reported intensity for each publicid.
- Validate the catalogue identifier and time columns, remove those that are missing, sort by time, and obtain distinct candidate identifiers.
- Extract latitude and longitude values.
- Bind the cell tables, select the required fields.
- Remove repeated *public_id + cell_id* pairs.
- Sort and then write output.

Design Choices & Limitations

- Aggregate reporting areas are returned, one MMI summary and one contributing count, not individual Felt RAPID observations. An aggregation method of ‘median’ was used when

8 https://arrow.apache.org/overview/

querying the API. GeoNet offers ‘max’ or ‘median’ aggregation, with max as the default[9]. Median was chosen so that the retrieved values represented the average of reports, rather than just the maximum value recorded which may be less representative of the overall experience of shaking at that reporting area.
- A pause of 0.2 seconds between requests, as well as a maximum of three attempts for each request and a 45 second timeout after an attempt, was used as best practice and to avoid being banned/throttled from the API.
- Empty results are retained as this is still a legitimate request outcome.
- Irregular JSON representations are processed consistently by handling missing or nested properties, flattening them into a sequence, and then taking the first value and trimming it[10]. They are then made numeric.
- The cell identifier is selected by API feature ID, if this is not available a geohash property is used, and if both are unavailable a generated value is used. A limitation is that the final fallback option is dependent on the ordering of features.
- The *count* value is rounded to avoid errors resulting from decimal point inclusions.
- A cache is implemented primarily to avoid having to re-request/download every earthquake within the date range each time the script is rerun, as the total process can take a very long time. A limit created by the cache is that successful responses are not automatically updated when new reports are added and/or the median is changed. A complete rerun from scratch ensures the data is up to date.
- Network errors and unsuccessful HTTP[11] responses are detected and retried after a delay, up to a limit of 3 times.
- Progress messaging is used primarily so the user can see how far along the process is.
- A safeguard was implemented wherein the script checks that every request completed successfully as either *success* or *no_reports*, and stops to notify the user. This is important as missing results due to a network error should not be counted the same as an actual lack of reports.
- Removal of duplicate *public_id + cell_id* pairs stops repeated features from multiplying a cells influence. However, as this is done with a deterministic rule of keeping the first occurrence, a limitation occurs if two rows with the same key have different values for intensity or count. This is a limitation that would require extra research and development to improve.

Script_03 – Moment Tensor

Output

An improved Earthquake Catalogue with moment-tensor Mw values, saved to *data_processed/geonet_catalogue_with_mw.parquet*.

Main Script Functions

- Download GeoNet’s moment-tensor solutions from their GitHub repository.
- Standardise naming and read columns as text (see Script_01).
- Select a single solution where more than one is available.
- Join the moment-tensor Mw to the catalogue by *publicid.*
- Write output parquet.

Design Choices and Limitations

- When *publicid* is missing, *event_id* is used in its place.

9 https://api.geonet.org.nz/#intensity
10 https://geojson.org/
11 https://developer.mozilla.org/en-US/docs/Web/HTTP/Reference/Status

- A solution is excluded if it has an empty identifier, the explicit placeholder value 9999999[12], a missing Mw, or an Mw < 0 or > 11[13].
- For each publicid, a candidate is selected based on highest variance reduction, then highest station count in the case of a tie, and finally by source row as a final tie breaker. Variance reduction was chosen from amongst the available options as it was the only real contender, with station count a clear second. A limitation is that no minimum is set for an acceptable variance reduction or station count, and so a low quality solution can be selected if it is the only one available or if all available solutions are poor. Another limitation is that the final deterministic tie-break depends exactly on source row order, and so is not reproducable.
- Fault information (strike, dip, rake, etc) was initially included, however as this was not used in later modelling, it was removed in the final version.
- Only the Mw is taken from the CMT file. Event time, latitude, longitude, and depth remain unchanged in the catalogue. It is important to note that the depth used is not the CMT depth.
- The GeoNet moment-tensor GitHub repository is described as "generally updated on an infrequent or irregular basis"[14]. This may lead to moment-tensor data being unavailable for more recent earthquakes. Moment-tensor information is also noted to generally only be held for earthquakes of magnitude 4 or greater[15].
- Events without a valid matched Mw are retained with Mw = NA, and removed in a later script.

Script_04 – Build Earthquake-Cell Dataset

Output

The earthquake dataset, saved to *data_processed/NZ_FR_IAR_dataset.parquet* (New Zealand Felt RAPID Intensity Attenuation Relation Dataset).

Main Script Functions

- Link earthquake events to their associated report cells.
- Calculate approximate epicentral and hypocentral distances.
- Apply validity rules.
- Write the earthquake dataset.

Design Choices and Limitations

- Event coordinates and event depth become *event_longitude*, *event_latitude*, and *event_depth_km*. This is to avoid mixing up event and report coordinates and depths.
- A minimum Mw filter of >=3.5 is applied.
- Some previous checks and filters are repeated here as an extra safeguard.
- A left join attaches event data to every cell observation, leading to multiple cells per event.
- Observations with no matching event/missing fields are removed from the dataset.
- A check is made that report and event coordinates are not missing and that they fall within the NZ bounding box coordinates (see Script_00). A check is made that event depth is not missing and is between 0 and 700 km[16]. Both of these are first-pass data checks that can be subsequently tightened during exploration and modelling.
- For *Repi_km*, rows with valid event and report coordinates are selected, and two spatial point collections are constructed. EPSG:4326[17] (WGS 84) is explicitly assigned to the coordinates. Spherical geometry[18] is used for the geographic coordinates to account for the curvature of the earth, and then distances between each report point and it's corresponding

12 https://github.com/GeoNet/data/blob/main/moment-tensor/README.md
13 https://www.geonet.org.nz/earthquake/intensity
14 https://github.com/GeoNet/data/blob/main/README.md
15 https://github.com/GeoNet/data/blob/main/moment-tensor/README.md
16 https://www.britannica.com/science/earthquake-geology/Shallow-intermediate-and-deep-foci
17 https://geodata.tools/blog/what-is-geojson
18 https://r-spatial.github.io/sf/reference/geos_measures.html

event point are calculated. This accounts for the curvature of the earth, and also avoids potential subtraction errors across the antimeridian[19].

- Rhypo_km is calculated using the formula *sqrt(Repi_km^2 + event_depth_km^2)*[20].

Script_05 – Vs30

Output

The final earthquake dataset for later analysis and modelling, saved to *data_processed/NZ_FR_IAR_final_dataset.parquet*.

Main Script Functions

- Extract Vs30 values at the report coordinates.
- Filter for plausibility.
- Write the final dataset.

Design Choices and Limitations

- The Foster et al. Vs30 raster file must be downloaded manually, the instructions for doing this are outlined in the script comments. This is because we initially had trouble locating the Vs30 from the FosterGeotech GitHub, and so were using a 2023 Vs30 raster from QuakeCore which had been developed from the FosterGeotech one. As a result, any file with the appropriate filename and filepath will be read as if it is the 2019 Foster et al. Vs30 raster, so it is important to verify that the correct file has been downloaded and used.
- As the same reporting coordinate can appear for many earthquakes, distinct valid latitude-longitude pairs are retained, sorted, and given a temporary identifier. This ensures that the site condition field changes by location and not arbitrarily by earthquake.
- EPSG:2193 (NZTM2000)[21] is assigned as per the FosterGeotech GitHub specifications[22]. Before this, the raster extent is transformed to EPSG:4326 and reporting coordinates are filtered to ensure they are within these bounds, in order to avoid trying to apply NZTM2000 which is intended for New Zealand to distant non-NZ coordinates.
- The coordinates are projected to the raster CRS in order to avoid reprojecting the raster and creating a new grid[23]. It also puts both the sites and raster into a shared metres based coordinate system.
- When a cell is missing from the Vs30 raster, a search is conducted for a valid cell within a 200 m radius to substitute. This is for practical purposes of recovering otherwise unavailable values, and does not reflect any claim that site conditions within 200 m of each other are the same. A value of 200 m was settled on as a trade-off between loss of data and cell-to-cell proximity. 1000 m retained 99.92% of observations, while 100m retained 94.51%, with the obvious advantage that cells this close are significantly more likely to have similar Vs30 values. 200 m retained 96.76% of observations, while still being a small enough distance to make similar Vs30 values between cells likely. Substituted and missing cells largely sit along the coast or the edges of lakes and rivers, where the aggregation of coordinates into reporting cells can lead to the central point used as the coordinate for that cell being in a body of water and thus having no valid Vs30 value.
- Vs30 values are filtered to keep only those between 98 and 2197 metres per second inclusive. Values outside that range become NA. The United States Geological Survey (USGS) uses this range[24]. Labelling as NA was done to keep otherwise valid earthquake-cell

19 https://www.rfc-editor.org/rfc/rfc7946.html#section-3.1.9
20 https://link.springer.com/article/10.1007/s10518-023-01822-8
21 https://www.linz.govt.nz/guidance/geodetic-system/coordinate-systems-used-new-zealand/projections/new-zealand-transverse-mercator-2000-nztm2000
22 https://github.com/fostergeotech/Vs30_NZ/blob/master/README.md
23 https://rspatial.github.io/terra/reference/project.html
24 https://www.usgs.gov/programs/earthquake-hazards/science/vs30-models-and-data

pairs. This was so that these could be used in later exploratory data analysis, and to not potentially introduce sampling bias by removing them before analysis.

Fields in the Final Dataset

Field	Type	Unit	Explanation
publicid	Character	None	GeoNet earthquake identifier
cell_id	Character	None	Reporting cell identifier within an event
origintime	UTC date-time	UTC	Event origin time
event_longitude	Double	Degrees	Epicentre longitude
event_latitude	Double	Degrees	Epicentre latitude
event_depth_km	Double	km	Focal depth (not CMT centroid depth)
report_longitude	Double	Degrees	Felt RAPID cell longitude
report_latitude	Double	Degrees	Felt RAPID cell latitude
reported_mmi	Double	MMI score	Median reported intensity in that event's cell
report_count	Integer	Reports	Total number of reports contributing to cell median
Mw	Double	Magnitude	Moment-tensor magnitude
Repi_km	Double	km	Spherical surface distance between event and cell
Rhypo_km	Double	km	Approximated hypocentral distance based on Repi_km and depth
Vs30_m_s	Double or Missing	m/s	Extracted mapped site estimate of Vs30

AI Use Acknowledgment

ChatGPT was used to assist in adding progress and warning messages to the pipeline scripts, providing feedback on where to add these and what checks were important. The prompt used was "Review this pipeline of scripts, and provide recommendations on where output messages can be added to provide notification to the user of successful/failed runs (and especially to notify the user of how far progressed downloads are), and other warnings, checks, and messages. Advise on how best to implement these.", along with the script 00-05 files.